A Unified 2D Framework for DeepLesion Detection, Segmentation and Short Report Generation

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ABSTRACT

In previous work, we integrated large language models (LLMs) into the lesion segmentation model based on the ULS23 DeepLesion dataset, using short-form findings from the reports. In this study, we developed a unified 2D lesion analysis framework that integrates LLM-based reasoning, lesion bounding box detection, segmentation, and radiology report generation from the original DeepLesion dataset. In the testing phase, we achieved relatively high lesion bounding box detection accuracy with mAP50 of 70.1%, mAP50-95 of 46.4%; Lesion segmentation performance with a Dice score of 62.6%; short report generation accuracy with BLEU_1 score of 64.3%, BLEU_4 score of 49.6%, METEOR of 34.7%, and ROUGE_L of 60.1%. In this work, we address the challenging issue of segmentation in the original DeepLesion dataset and achieve a 28.5% Dice score improvement over the nnUNet lesion segmentation model. We also integrated spatial and anatomical context into the DeepLesion short report generation. We released the implementation, dataset, and models on Github. https://github.com/ruida/2D_DeepLesion_Foundation

Keywords: LLM, Foundation model, DeepLesion, detection, segmentation, report generation

1. INTRODUCTION

Lesion detection, segmentation, and report generation play an essential role in medical analysis and clinical diagnosis. Recent developments in large language models (LLMs) and vision-language foundation models have created an opportunity to develop a general-purpose computer-aided diagnosis (CAD) system for medical image analysis and to enable several essential downstream clinical workflows, such as lesion detection to localize lesion regions, lesion segmentation to generate the binary lesion mask or delineate lesion boundaries for area- and volume-based measurements, and short report generation to provide a briefing on findings and impressions. However, previous studies [1, 2, 3, 4] primarily focused on lesion detection with CNN-based models and used one-hot encoding and decoding for short report generation. DeepLesion dataset-based detection and segmentation remain challenging tasks due to 1) the limited lesion annotation, 2) the spatial imbalance for tiny lesions, and 3) the scattered lesion locations in the 512x512 CT image slices that are hard to identify in both detection and segmentation tasks. From the radiologist's perspective, the clinical lesion assessment, localization, delineation, and description should follow a sequential and interconnected workflow, yet most existing studies treat them as separate independent tasks. We proposed a unified 2D DeepLesion framework that integrates anatomical and semantic context into lesion detection, segmentation, and short report generation from the original DeepLesion dataset.

2. METHODS

Overall architecture

Given a lesion image, the proposed unified 2D framework aims to localize the lesion with a bounding box, segment the lesion, and generate a lesion-specific short report. Figure 1 illustrates the overall workflow and its main components.

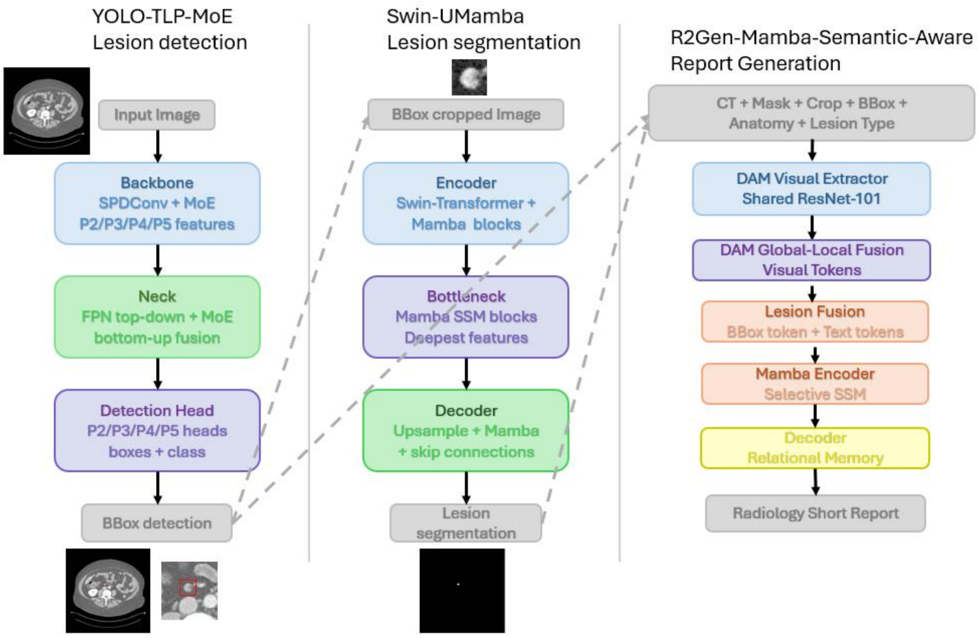


Figure 1. Overall architecture

Lesion Detection

In the DeepLesion dataset, detecting tiny lesions is challenging. Most lesions occupy a small fraction of pixels in a single 512x512 high-resolution image slice, resulting in spatial imbalance. Conventional stride convolution and pooling operations in CNNs can progressively reduce spatial resolution and consequently suppress or eliminate the lesion features during extraction in the downsampling path. For lesion detection, preserving fine-grained lesion representation with broader anatomical context appears to be a challenging task. We are motivated to develop a hybrid detection architecture that combines three complementary design building blocks: lossless spatial downsampling (SPDConv from YOLO-TLP [5]), adaptive feature routing (Mixture-of-Experts, MOE, from YOLO-Master [6]), and multi-scale global attention (PSA-SPPF from YOLO-TLP).

Lesion Segmentation

Applying segmentation directly to the original 512x512 high-resolution DeepLesion dataset is challenging. Most lesion regions are small and occupy only a small fraction of the image spatially. The spatial imbalance results in a significant degradation in segmentation performance with conventional medical segmentation models. For example, nnUNet's [7] segmentation based on the original DeepLesion dataset yields only a 34% mean Dice score in the test phase, making it not an ideal solution for DeepLesion segmentation. Most likely, the tiny regions were suppressed during the downsampling step in nnUNet. Also, the lesions are scattered around the 512x512 image slice, making segmentation more challenging. To address the issue, we motivated to applying lesion detection first to generate highly likelihood lesion regions. Then we segment the cropped legion region based on the detection results. We utilize the Swin-UMamba [8] to segment the cropped lesion regions. Based on the original DeepLesion dataset bounding boxes and the grab-cut-generated lesion labels, we crop the lesion region for both images and labels. Those cropped image-label pairs vary in image size. After generating the prediction labels for the cropped lesion regions, we merge or patch the labels back into the original 512x512 label images for evaluation.

Short Report Generation

Based on the detected lesion bounding box, we integrate the R2Gen-Mamba [9] architecture with the Describe Anything Model (DAM) [10] and a Bounding Box (BBox) anatomy-aware, lesion-type-aware architecture to generate the lesion short report. The original R2Gen-Mamba [9] generates the medical report from the entire 2D image and relies fully on visual features. This design ignores the related structured clinical metadata. In the DeepLesion dataset, each lesion can be roughly categorized into one of the 11 anatomical regions and one of the 10 lesion types, each contributing specific semantic cues to short report generation.

We propose the R2Gen-Mamba-Semantic-Aware framework (Figure 2), which jointly models lesion-focused visual representation, anatomical location, and lesion type to generate the short report. Global and local visual features are integrated through gated global-to-local fusion (attention) to produce lesion-aware DAM visual tokens. In parallel, the normalized bounding box coordinates are encoded as a spatial feature token, while the combined anatomy and lesion type text are encoded as semantic tokens. LesionFusion combines the DAM visual tokens, bounding-box token, and semantic text tokens into a conditioned visual sequence. The sequence is then processed by the Mamba selective state-space encoder and the relation-memory decoder to generate the final short report. Thus, the anatomy and lesion types provide the rough semantic conditioning for short report generation.

After detecting the bounding box from the YOLO-TLP-MOE model, we aligned the BBox with the pre-segmented anatomical region label to identify the rough anatomical region. During inference time, the BBox, rough anatomical region, and lesion type have been encoded as spatial and semantic context, which is hard to recover from images alone (i.e., the original R2Gen-Mamba [9]) or, particularly, for small or ambiguous lesions.

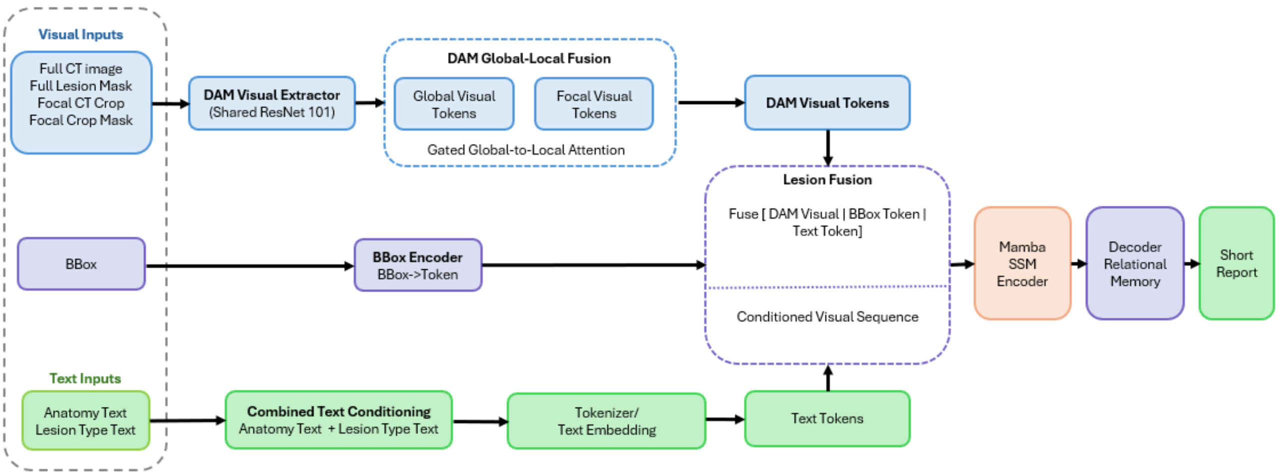


Figure 2. R2Gen-Mamba-Semantic-Aware framework: DAM + Anatomy + Lesion Type guided short report generation

To align the detection bounding box with the anatomical region, we used TotalSegmentator [11] to generate 3D segmentation maps of rough anatomical structures, then extracted the corresponding 2D slice from each 3D segmentation to match the DeepLesion 2D slice representation. Then, we derive an algorithm to allocate the detected bounding box coordinates to approximate organ regions, e.g., lung or liver. Figure 3 demonstrates the 2D image slice with detected bounding boxes and the corresponding segmentation map. The pseudo-code (Algorithm 1 in Appendix) explains the algorithm to aligning the bounding box with a rough anatomical region, then mapping it to the 11 anatomy regions.

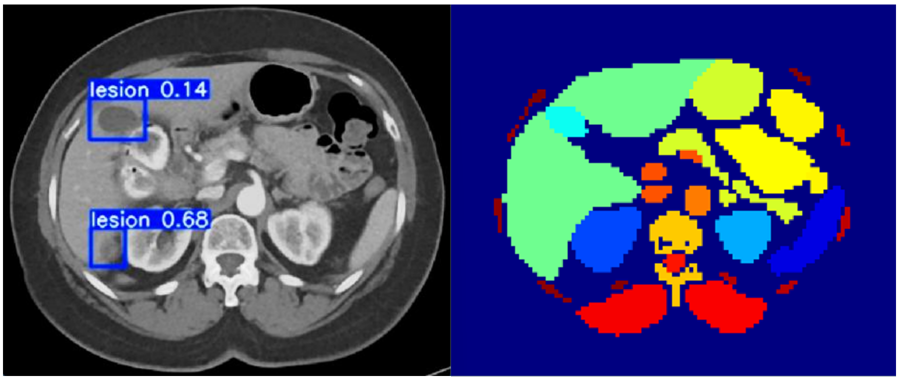


Figure 3. Align bounding box with the rough anatomical region ID

During inference, the lesion bounding box predicted by YOLO-TLP-MOE is used to define the focal lesion crop and localize the lesion within the anatomical segmentation map. The detected bounding box is aligned with the corresponding 2D TotalSegmentator label map, and the anatomical structure with the greatest overlap inside the box is mapped to one of the predefined rough anatomical regions. The lesion type is predicted from the focal lesion region using a lesion-type classification module. The resulting anatomy description and lesion type are converted into short-text phrases, such as "lung" and "nodule," and combined into a single conditioning sequence. The text sequence, joined with DAM visual tokens, is used to generate the final short report.

3. RESULTS

For the lesion Bounding Box (BBox) Detection module, we compared the proposed YOLO-TLP-MOE models with three baseline models, YOLO-TLP [5],YOLO-Master (MOE) [6], and YOLO-12 [12]. We utilize the standard YOLO detection metrics: mAP50, mAP50:95, Precision, and Recall comparing the detection performance (Table 1). Figure 4 demonstrates a few comparison cases for lesion detection.

Table 1. Comparison of bounding box detection performance for DeepLesion dataset

	Precision	Recall	mAP50	mAP50:95
YOLO-12 [12]	0.570	0.514	0.505	0.291
YOLO-TLP [5]	0.659	0.640	0.673	0.442
YOLO-Master (MOE) [6]	0.660	0.643	0.672	0.422
YOLO-TLP-MOE	**0.665**	**0.674**	**0.701**	**0.464**

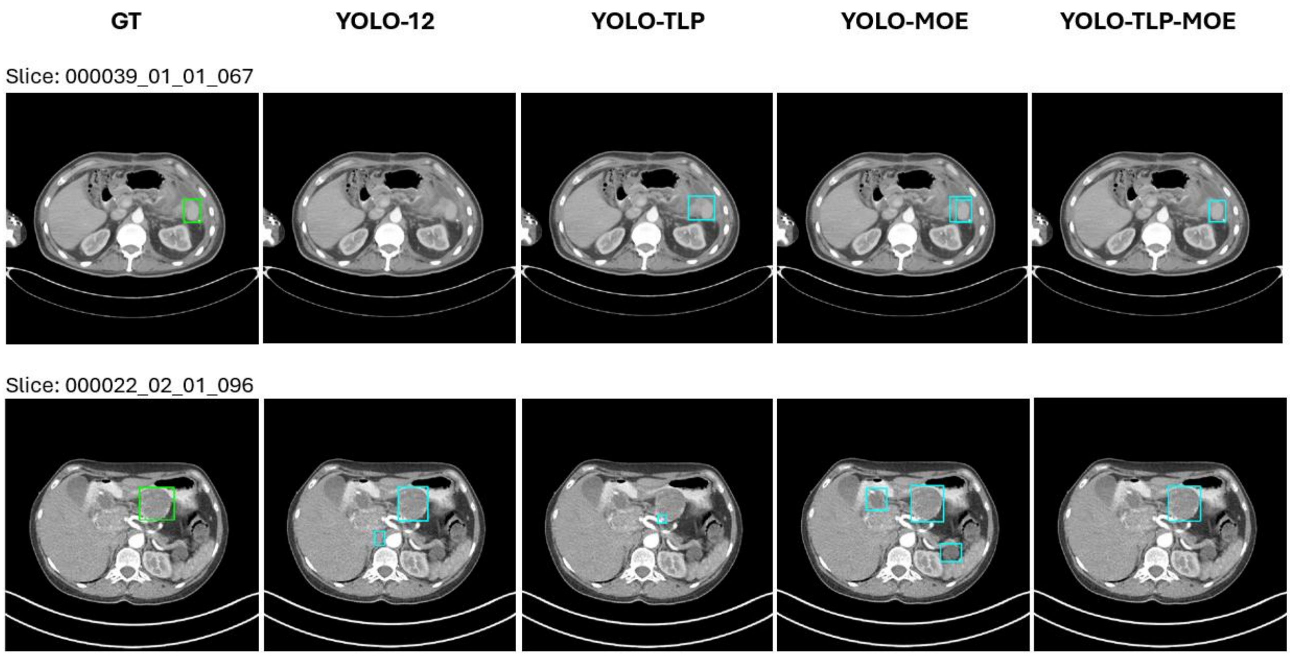
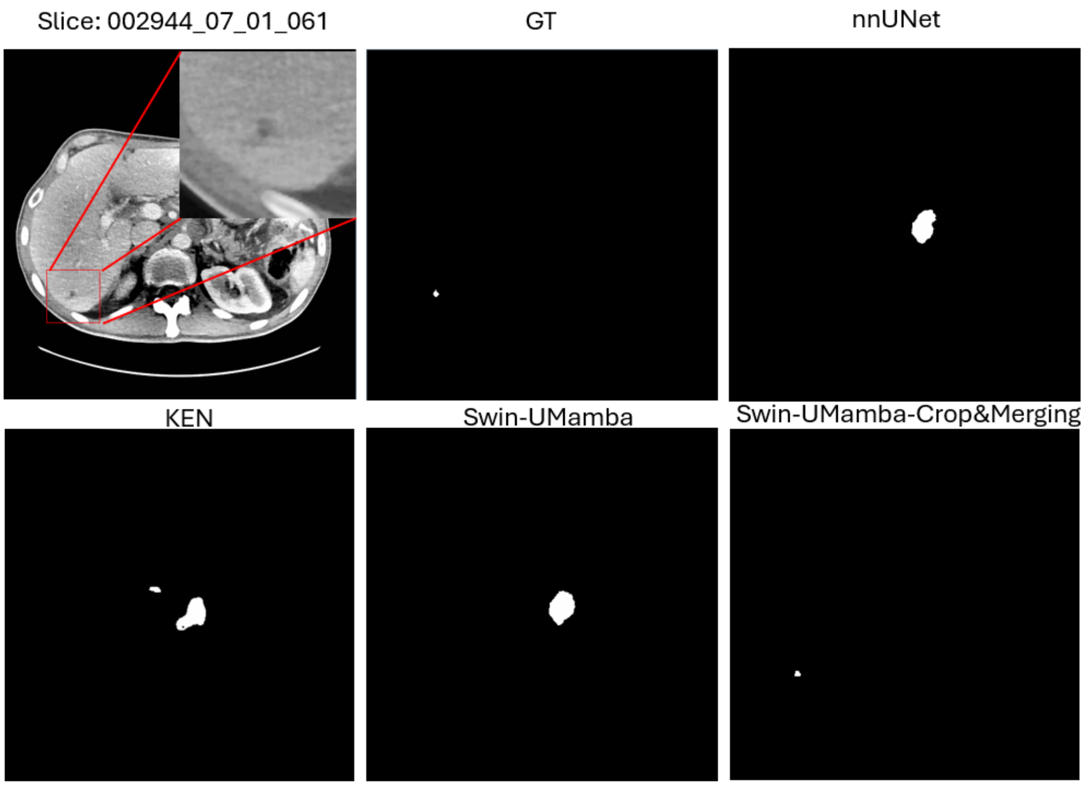


Figure 4. Comparison of predicted lesion bounding box detection

Based on the YOLO-TLP-MOE detected bounding boxes during the testing phase, we test the Swin-UMamba segmentation model on the cropped lesion regions and merge or patch the cropped segmentation map back into the original 512x512 image space. We compare segmentation performance (Table 2) with nnUNet, Swin-UMamba, and KEN using the mean Dice score—all the three models apply direct segmentation to the original DeepLesion dataset. Figure 5 shows the visual comparison of lesion segmentation results.

Table 2. Comparison of lesion segmentation performance for DeepLesion dataset

	Mean Dice	Hausdorff95	Mean IoU
nnUNet [7]	0.341±0.413	55.711±70.438	0.298±0.377
KEN [13]	0.529±0.002	224.144±1.752	0.507±0.001
Swin-UMamba [8]	0.401±0.426	57.244±74.119	0.356±0.397
Swin-UMamba crop & merging	**0.626±0.416**	**39.036±65.973**	**0.576±0.397**


Slice: 002944_07_01_061
GT
nnUNet
KEN
Swin-UMamba
Swin-UMamba-Crop&Merging

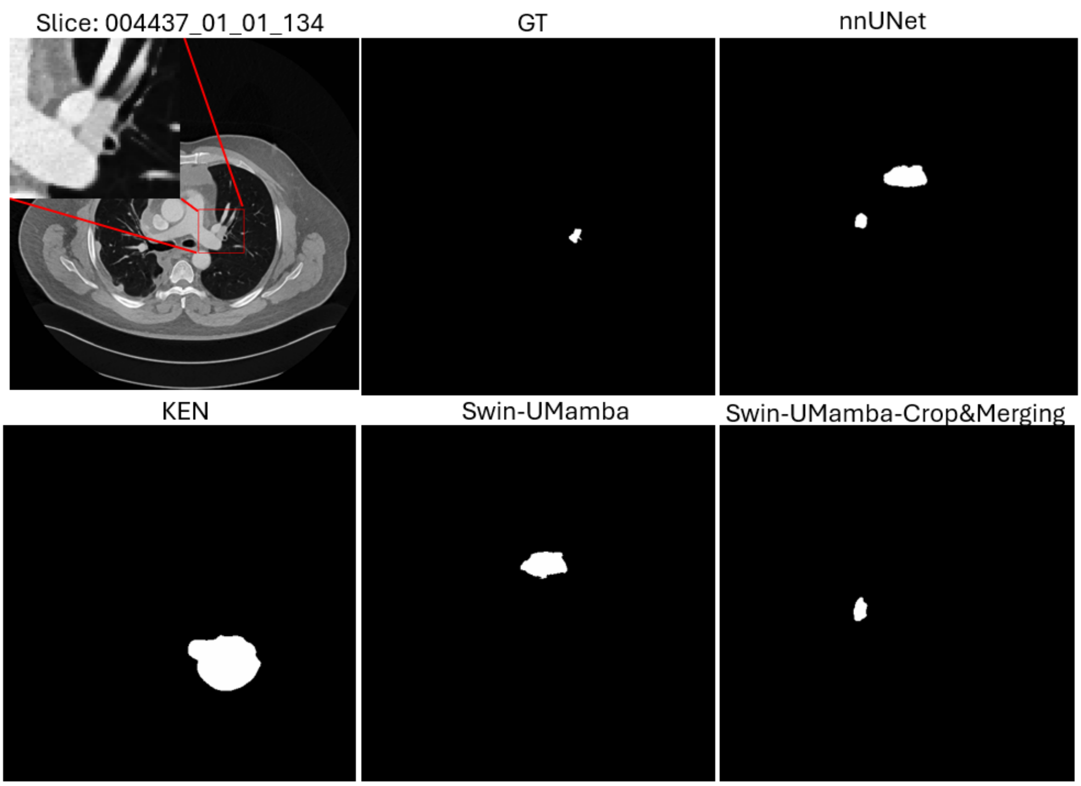

Slice: 004437_01_01_134
GT
nnUNet
KEN
Swin-UMamba
Swin-UMamba-Crop&Merging

Figure 5. Qualitative comparison of the predicted lesion segmentation results. The ground truth segmentation is followed by each model's segmentation result (Table 2). The GT represents the ground truth. Slice 004437_01_01_134 represents the missing segmentation case.

The R2Gen-Mamba-Anatomy-Aware model adds anatomy term, lesion type and the detected bounding box tokens to the R2Gen-Mamba architecture's encoder to improve lesion short report generation. To evaluate the performance of short report generation, we compare our proposed model with the latest state-of-the-art medical report generation models, MedGemma 1.5 and Qwen3-VL. As shown in Table 3, our proposed R2Gen-Mamba-Anatomy-Aware model outperforms MedGemma and Qwen3-VL by a large margin. We fine-tuned the MedGemma and Qwen3-VL models using the same train, validation, and test datasets of the DeepLesion. We implemented the Anatomy-Aware mechanism into MedGemma 1.5 and Qwen3-VL encoders and conduct the ablation experiments with Anatomy-Aware MedGemma 1.5 and Qwen3-VL. Figure 6 compares a few cases for the short report generation in testing phase.

Table 3. Comparison of short report generation performance for DeepLesion dataset

	BLEU_1	BLEU_4	METEOR	ROUGE_L	Precision	Recall	F1
MedGemma 1.5 [14]	0.334	0.181	0.219	0.367	0.351	0.441	0.370
Qwen3-VL [15]	0.344	0.156	0.353	0.398	0.435	0.451	0.413
R2Gen-Mamba [9]	0.563	0.307	0.251	0.545	0.651	0.602	0.595
MedGemma1.5-Anatomy-Aware	0.447	0.297	0.344	0.472	0.497	0.607	0.520
Qwen3-VL-Anatomy-Aware	0.391	0.170	**0.407**	0.443	0.469	0.526	0.465
R2Gen-Mamba-Anatomy-Aware	**0.643**	**0.496**	0.347	**0.601**	**0.704**	**0.666**	**0.650**

Figure 6. Comparison of the short report generation

4. CONCLUSION

We proposed an integrated DeepLesion 2D framework for tiny-lesion detection, detection-assisted segmentation, and lesion focus short-report synthesis from the original-resolution DeepLesion 2D CT image slices. The proposed YOLO-TLP-MOE detector was capable of retaining fine spatial detail information in layer-wise feature propagation along with multiscale and mixture-of-experts routing. The search-and-segment strategy depended on the predicted lesion bounding boxes to work around the tiny-lesion spatial imbalance issue and advance lesion delineation. The report-generation module was motivated by an anatomy- and semantically aware mechanism to combine global image features, localized lesion features, bounding-box information, coarse anatomical region, and lesion type to generate semantically grounded short reports. The proposed framework achieved an mAP50 of 0.701 for lesion detection, a mean Dice score of 62.6% for lesion segmentation, and BLEU-1 and BLEU-4 scores of 0.643 and 0.496 for report generation. These results support the feasibility of effectively downstreaming lesion localization, delineation, and reporting tasks within a unified framework and suggest that spatially focused, semantically conditioned modeling can improve automated lesion report generation.

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